Q.28 Define refrigeration. Derive the expression for coefficient of performance of refrigeration.

Q.29 Discuss in brief the properties of refrigerant.

Q.30 Differentiate between Isothermal and adiabatic process.

Q.31 Explain the significance of Vander wall's equation.

Q.32 Discuss in brief the joules experiment.

Q.33 Prove that for an isometric process, du = Cvdt.

Q.34 Explain in brief system, Surrounding and universe.

Q.35 Explain in brief Isometric and Isobaric process.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain in detail vapour compression refrigeration cycle with the help of neat diagram.

Q.37 Write short notes on any two of following.

a) State function and path function

b) Second law of thermodynamics

c) Entropy change for reversible process.

d) Internal energy for the system

Q.38 Define Isothermal process. Prove that work done for an ideal gas in Isothermal process is given by $Q=W=RT\ln(P_1/P_2)$.

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Roll No. ....

4th Sem / Chem, Chem Engg. (Spl. Paint Tech), Chem Engg. (Spl. Polymer Engg.)

Subject:-Chemical Engineering Thermodynamics / Engg. Thermody.

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Entropy is a measure of the _____ of a system.

a) Disorder b) Orderly behavior

c) Temperature changes d) None of these

Q.2 Isobaric process means a constant ____ process.

a) Temperature b) Pressure

c) Volume d) entropy

Q.3 Which of the following is path function?

a) Enthalpy b) Entropy

c) Internal Energy d) Work

Q.4 Which of the following is intensive property?

a) Enthalpy b) Temperature

c) Entropy d) Internal energy

Q.5 A system which interact with surrounding by exchange of both mass and energy (Heat) is called

a) Open system b) closed system

c) Isolated system d) None of these

Q.6 At absolute zero temperature the entropy of pure crystalline substance is zero, this is

a) Maxwell's equation

b) Nernest equation

c) Third law of thermodynamics

d) Second law of thermodynamics

Q.7 Efficiency of heat engine is given by

a) Work output / heat input

b) Heat input / Work output

c) Heat input *Work output

d) None of these

Q.8 Which of the following expression is true?

a) H=U-PV b) H=U+PV

c) H=V+PU d) H=U-RT

Q.9 Thermodynamics is the study of energy in the form of

a) Heat b) Work

c) Both (a) and (b) d) None of these

Q.10 Enthalpy of an ideal gas depends upon.

a) Pressure b) Temperature

c) Volume d) All of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define heterogeneous system.

Q.12 Define Gibbs free energy.

Q.13 Define Amagats's law.

Q.14 Write formula for coefficient of performance.

Q.15 Define polytropic process.

Q.16 Define extensive properties.

Q.17 Define heat pump.

Q.18 Define irreversible process.

Q.19 Write any two commonly used refrigerants.

Q.20 Mention the absolute scale of temperature measurement.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain in brief the zeroth law of thermodynamics.

Q.22 Explain in brief state function and path function along with examples.

Q.23 Explain in brief the first law of thermodynamics for closed system.

Q.24 Derive the expression for work done for ideal gas undergoing reversible Isobaric process.

Q.25 Explain in brief the Carnot cycle with the help of neat diagram.

Q.26 Explain in brief thermodynamics temperature scales.

Q.27 State and explain the third law of thermodynamics.